

flea NEWS

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Department of Entomology Iowa State University Ames, Iowa 50011-3222

ANNOUNCEMENTS

As of 4-IV-1991, Mrs. Annette K. Secombe transferred from the British Museum (Natural History) to a position elsewhere in the Civil Service. She has been replaced by Miss T. M. Howard who is now responsible for the flea collection. Requests and correspondence concerning the flea collection should be addressed to her in care of the Department of Entomology.

The Ceratophyllidae Revisited

We recently received the following commentary from the Honourable Miriam Rothschild, FRS, with a request that it be included in Flea News.

"It is most fortunate that R. E. Lewis has published such a useful volume covering the currently accepted valid taxa of the Ceratophyllidae; apart from its intrinsic excellence it compliments and enhances our volume cataloguing the Rothschild and British Museum Collection of this 'difficult' group of fleas.

"In this we have described and illustrated the genera and subgenera of the family (205 text figures, 532 photographs, and 48 distribution maps), and in addition provided 175 maps showing the geographical distribution of all the species and major subspecies of the family and their hosts.

"As circumstances prevented us describing the taxonomy of the fleas down to species level (as in previous volumes) we did

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not designate the monograph as No. 8 of the *Catalogue of the Rothschild Collection of Fleas*, but it does belong to that series, and we thus fulfilled on behalf of the Trustees of the British Museum (Natural History) the terms of the N. C. Rothschild gift (*The Rothschild Collection of Fleas*, Vol. I, 1953, pp. 357-358)." M. Rothschild, 9-I-1991.

The Society of Vector Ecology

SOVE is an organization of professionals dealing with ecology, prevention, and control of insects and other arthropods and organisms of public health importance. The Society holds an annual conference as well as regional meetings for the purpose of sharing information among its membership. This year's annual conference will be held November 18-20 at the Nugget Hotel, Sparks, Nevada. Contact B. Fred Beams, P. O. Box 87, Santa Ana, California 92702, Ph. (714) 971-2421 or Fax No. (714) 971-3940.

J U L Y

1991

Announcements (Continued)

Drs. Jay and Marion Georgi send the following news about their activities:

"We are currently conducting personally funded research on artificially fed *Ctenocephalides felis* with the goals of improving the efficiency of our methods, determining biologic optima, maximizing production, and developing research and commercial applications. Our present 'artificial dog' has 20 units, each populated with 100-200 fleas and daily yields 3000 to 4000 adult fleas. Many more fertile eggs than this are produced but we have yet to determine optimum cultural conditions for very large numbers of larvae. We would be happy to supply information to others interested in adopting our techniques and to exchange information with those already engaged in the artificial rearing of fleas.

"Having completed the manuscript of *Canine Clinical Parasitology*, to be published later this year by Lea & Febiger of Philadelphia, we are now starting to work on *The Cat Flea and the Artificial Dog*, which we intend will include both classic information and the results of current research on the biology of *Ctenocephalides felis*, including those stemming from our own investigations alluded to above. Although we are fortunate to have the excellent library facilities of Cornell University at our disposal, we would nevertheless be most grateful for reprints of published reports on any subject directly related to the cat flea."

Drs. Georgi may be contacted at: *Fleadata*, 132 Starr Stanton Road, Freeville, New York 13068-9802. (607) 844-8507.

We have recently been apprised of the deaths of **Dr. Renata Costantini** (29-I-1991) and **Prof. Dr. med. Heinz Eberhard Krampitz** (2-VII-1921 - 12-V-1991). We hope to have biographical information on these two colleagues for the next issue of *Flea News*.

In her communication about Dr. Costantini, Dr. Albertina Iori of the Istituto di Parassitologia, Università di Roma, included an inventory of the flea collection at that institution. Interested parties may obtain a copy by request to *Flea News*.

Your attention is drawn to a new newsletter called **Plague Surveillance** produced by the Division of Vector-Borne Infectious Diseases, Center for Disease Control, P. O. Box 2087, Fort Collins, CO 80522 and edited by Drs. Allan Barnes and David Dennis. The introduction to volume 1, number 1

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(June, 1991) states "This issue...is the first in a continuing series of communications on plague epidemiology, surveillance and control in the western United States. The series will emphasize data on trends among animal vector and pathogen populations involved in the natural plague cycle and present summaries of human cases, epizootic investigations, surveillance methodologies, preventive control, and other items concerned with plague. It may, from time-to-time, present information on international plague events."

BOOK REVIEWS

Opredelitel' blokh Mongol'skoi Narodnoi Respubliki. [Keys to the fleas of the Mongolian National Republic.] By Goncharov, A. I., T. P. Romashcheva, B. K. Kotti, A. Bavaasan & S. Zhigméd. 1989. Ulan Bator. 415 pp., 426 figs. 6 rubles.

The flea fauna of Mongolia has attracted the attention of pulicologists since Jordan (1929) reported on collections made in the area by H. M. Jettmar in 1927-1928. In the past few decades the country has been the focus of cooperative field studies between Mongolian scientists and workers from Hungary, Czechoslovakia, Germany and the USSR. This volume is a summary of the accumulated information about the flea fauna of the Republic.

Following a brief introduction is a section dealing with the collection of fleas and their preparation for study. This is followed by a section treating the anatomy of adult fleas, with particular emphasis on structures of taxonomic importance. The bulk of the book is devoted to keys to the 190 species and subspecies now known to occur within the political boundaries of the country. The book concludes with a list of the 426 illustrations and their sources, a list of the taxa and their figure numbers, a list of junior synonyms, a bibliography, an index and a table of contents.

The format is typical of most Russian catalogues published during the last 50 years in that each taxon listing is followed by a brief statement about the distribution and host associations. Additional references to the species are also usually cited.

In a work of this scope one must expect the introduction of nomenclatural variations that may not be in accord with commonly

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accepted current usage. Following is a review of the variants noted by the reviewer.

The authors follow the classification of Smit (1982) in splitting the Ctenophthalmidae from the Hystrichopsyllidae, but they retain the Ceratophyllidae and Leptopsyllidae as full families. The taxa are distributed as follows in the order in which they are treated. Numbers of species and subspecies are included along with the number of genera in parentheses. Pulicidae 11, (6); Ceratophyllidae 55, (11); Leptopsyllidae 66, (9); Ischnopsyllidae 4, (2); Ctenophthalmidae 42, (8); Hystrichopsyllidae 2, (1); Coptopsyllidae 2, (1) and Vermipsyllidae 8, (3).

Subspecies of *Xenopsylla conformis* (Wagner, 1903) are not recognized.

Oropsylla ilovaiskii Wagner & Ioff, 1926, is treated as a subspecies of *O. idahoensis* (Baker, 1904).

No subspecies of either *Amphalius clarus* (Jordan & Rothschild, 1922) or *A. runatus* (Jordan & Rothschild, 1923) are recognized.

Nosopsyllus laeviceps gobiensis Cyprich, Kiefer & Cendsuren, 1977, is kept separate from *N. l. ellobii* (Wagner, 1933) although it was synonymized by Smit (1980c: 78).

Citellophilus altaicus (Ioff, 1936) is treated as a subspecies of *C. tesquorum*. *C. sungaris* (Jordan, 1929) is also assigned as a subspecies of *C. tesquorum*.

No subspecies of either *Ceratophyllus enefdei* Ioff, 1950 or *C. rossittensis* Dampf, 1913 are recognized.

Ceratophyllus vagabundus dimi Mikulin, 1958 is retained, although it was synonymized with *C. v. insularis* Rothschild, 1906 by Brinck-Lindroth & Smit, 1971.

Ceratophyllus styx avicitelli Ioff, 1946 is retained, although synonymized by Liu *et al.*, 1986.

Igioffius taiganus (Scalon, 1950) is still assigned to the genus *Megabothris*.

Two undescribed subspecies of *Amalaraeus dissimilis* are mentioned.

Ochotonobius hirticrus (Jordan & Rothschild, 1923) is still assigned to the genus *Ctenophyllus*.

Frontopsylla (*Frontopsylla*) *luculenta parilis* Jordan, 1929 is retained, although synonymized with *F. (F.) luculenta* (Jordan & Rothschild, 1923) by Liu *et al.*, 1986.

Frontopsylla (*Frontopsylla*) *elata caucasica* Ioff & Argyropulo, 1934 is elevated to full species rank. Also, an undescribed subspecies of *F. elata* is

mentioned.

Frontopsylla (Frontopsylla) postprojicia Quan, Ai-min & Zhong-yuan, 1985. [The citation for this taxon does not appear in the reference section but the species obviously relates to: Liu Quan, Qi Ai-min & Li Zhong-yuan, 1985. *Acta zootax. sin.* 10(4): 431-433, figs. 1-6.]

Frontopsylla (Orfrontia) lapponica prior Scalon, 1965 is retained in figure captions 189-190, although the subspecies was synonymized with *F. lapponica* (Nordberg, 1935) by Smit & Wright, 1978a: 40.

Ophthalmopsylla (Ophthalmopsylla) extrema Ioff & Scalon, 1953 is treated as a subspecies of *O. (O.) volgensis*.

Ophthalmopsylla (Ophthalmopsylla) praefecta lenta Smit, 1967 is synonymized with *O. (O.) praefecta ecphora* Labunets, 1961.

Paradoxopsyllus alatau Shvartz, 1953 and *P. kalabukhovi* Labunets, 1961 are treated as subspecies of *P. hesperius* Ioff, 1946.

Paradoxopsyllus socrati Kunitskaya & Kunitsky, 1978 is treated as a synonym of *P. conveniens* Wagner, 1930.

Mesopsylla eucta shikho Ioff, 1950 is assigned as a subspecies of *M. tuschkan*.

Subspecies of *Peromyscopsylla bidentata* (Kolenati, 1863) are not recognized.

Ctenophthalmus pisticus pisticus Jordan & Rothschild, 1921 and *C. p. pacificus* Ioff & Scalon, 1950 are assigned to the subgenus *Asioctenophthalmus* Rostigayev, 1971.

Undescribed subspecies of *Neopsylla mana* Wagner, 1927 are mentioned.

Subspecies of *Neopsylla setosa* Wagner, 1898 are not recognized.

Subgenera *Hystrichopsylla* and *Hystroceras* are retained in the genus *Hystrichopsylla*.

With the addition of this and the following volume to the monographs by Ioff & Scalon, 1954, Sakaguti, 1962, Ioff, Mikulin & Scalon, 1965, Liu *et al.*, 1981 and Liu *et al.*, 1986 our knowledge of the flea fauna of eastern Asia nearly rivals that of the fauna of western Europe. -REL

The flea fauna of Xinjiang. By Yu Xin, Ye Rui-yu & Xie Xing-chu. Xinjiang People's Publishing House. 1990. Urumqi, China. 542 pp., 549 figs., pls. I-X. U.S. \$50.00 [includes air mail postage].

That this and the preceding volume should appear so recently is an interesting coincidence. The two areas are similar in size (Xinjiang = 635,800 sq. mi., Mongolia = 604,250 sq. mi.), topography, and, to a large

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extent, climate. However, the flea fauna of Xinjiang is somewhat poorer, currently consisting of only 181 named taxa in 40 genera, distributed as follows: Ceratophyllidae 50, (12); Coptopsyllidae 2, (1); Hystrichopsyllidae 35, (9); Ischnopsyllidae 3, (1); Leptopsyllidae 68, (8); Pulicidae 14, (6) and Vermipsyllidae 9, (3). The authors do not separate the Ctenophthalmidae from the Hystrichopsyllidae. Both areas reside in the Mongolian faunal region as delineated by Stegmann (1938) [*Principes généraux des subdivisions ornithogéographiques de la région paléarctique*. Faune de l'URSS. Acad Sci. URSS 1(2): 156], and the fauna is typical of the Palaearctic Region.

The book is divided into six parts. Part one consists of a general account of the order Siphonaptera with particular reference to Xinjiang Province and treats the history of flea research, the anatomy and morphology of both the adults and larvae as well as their biology, ecology, phylogeny and geographic distribution. The relation of fleas to disease and survey, preservation and control techniques are also included in this section.

The bulk of the book deals with systematic accounts of the indigenous species and their morphology, biology, ecology and medical significance, if known. While the species are illustrated by adequate figures, the captions are in Chinese only, and a reader unfamiliar with this language must refer to the species accounts to determine which figures apply to which taxon.

Part three is a brief English resumé of the book including an English summary of the six new taxa described in the text.

Part four contains three appendices dealing with; 1) a distribution table of the fleas of Xinjiang; 2) a list of Chinese names of the fleas of Xinjiang and their Latin equivalents; and 3) a list of Chinese names of host animals and their Latin equivalents.

Part five contains the literature cited arranged in Chinese, English and Russian.

Part six contains indices in both Chinese and Latin and a final appendix listing four additional taxa discovered after the work was completed. Following are the data pertaining to the six new species and subspecies.

***Coptopsylla lamellifer tarimensis* Yu, Ye & Cao, 1990.**

PRC, Xinjiang, Hetian district and bank of the Tarim River, from *Meriones meridianus* and *Brachiones przewalskii*, IX.1972 & XI.1987. Holotype male, allotype female, 11 male and 18 female paratypes.

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Vermipsylla yuae Yu & Li, 1990.

PRC, Xinjiang, Yili district, from *Ovis aries* and *Equus caballus*, XI.1985. Holotype male, allotype female, 1 male and 6 female paratypes.

Stenoponia meridiana Cao, Xie & Yu, 1990.

PRC, Xinjiang, S of the Talimu Basin, from *Brachyones przewalskii* and *Meriones meridianus*, XI.1987 & X.1979. Holotype male, allotype female. One female paratype from east of the Talimu Basin, from *M. meridianus*, IX.1987.

Ctenophyllus (Ctenophyllus) armatus altaicus Yu, 1990.

PRC, Xinjiang, Altai Mts, from *Ochotona alpina*, V-VI.1965-1986. Holotype male, allotype female, 9 male and 26 female paratypes.

Ophthalmopsylla (Ophthalmopsylla) volgensis wugiaensis Yu, Shen & Ye, 1990.

PRC, Xinjiang, Wuqia [=Wugia, Wukia, Uluggat, Ulugh Chat], 39.50N 74.20E, ca. 2050m, from *Allactaga sibirica*, VII-VIII.1986. Holotype male, 8 male paratypes.

Callopsylla (Callopsylla) yui Ye & Jiang, 1990.

PRC, Xinjiang, Qitai county, Beita Mt, from *Lagurus lagurus*, V.1980. Holotype male, allotype female, 20 male and 21 female paratypes.

As far as we can ascertain all primary types (i.e., holotypes) are deposited in the Xinjiang Institute for Endemic Disease Control and Research, Urumqi. -REL

REPRINTS RECEIVED

The following readers have sent reprints or other literature sources since Flea News 41. Thank you. We are most grateful for your cooperation! J.-C. Beaucournu, V. Czerny, D. Cyprich, A. Dudich, L. Durden, T. Galloway, R. George, A. I. Goncharov, Gong Z.-d., M. Greenwood, M. Krumpál, Li K.-c., Qi Y.-m., H. Ribeiro, P. Sleeman, M. Stanko, Xie B.-q., Ye R.-y. and Yu X.

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MAILING LISTS

Additions

Dr. Harry N. Cunningham, Jr., Division of Science, Engineering & Technology, Penn State Erie, Station Road, Erie, PA 16563-0203.

Ms. Theresa M. Howard, Siphonaptera Section, British Museum (Natural History), Cromwell Road, London SW7 5BD, England.

Dr. Albertina Iori, Istituto di Parassitologia, Università "la Sapienza", Piazzale Aldo Moro, 5, 00185 Roma, Italia.

Dr. Susan Lewis, Medical Entomology Centre, Austin Building, New Museums Site, Pembroke Street, Cambridge CB2 3DX, England.

Dr. James D. Long, V.S.C.D., E.H.S., Department of Public Health Services, P. O. Box 85261, San Diego, CA 92186-5261.

Dr. Donald H. Miller, Department of Science, Lyndon State College, Lyndonville, VT 05851.

Mr. Vernon A. Nelson, Department of Biology, Southern Connecticut State University, New Haven, CT 16515.

Ms. Monique Palaysi, Service de Documentation, Rhône Mérieux, 1, Chemin du Calquet, 31057 Toulouse Cédex, France.

Dr. Robert M. Timm, Museum of Natural History, University of Kansas, Lawrence, KS 66045.

Changes

Mr. William Bender, 3712 Sachem Hill, Cincinnati, OH 45226 USA

Dr. Jenella Loye, Department of Entomology, University of California - Davis, Davis, CA 95616.

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R. Mulcahy, 7 Oxford Street, Lancaster LA1 2NF, England.

Provisional lists of taxa described since 1989

1989

Palaeopsylla laxidigita Xie & Gong
Palaeopsylla polypina Xie & Gong
Palaeopsylla medimina Xie & Gong
Ctenophthalmus (Palaeoctenophthalmus) harputus Aktaş
Ctenophthalmus (Ctenophthalmus) solutus nasutus Beaucournu, Valle & Launay
Citellophilus simplex rosickyi Cyprich
Xenopsylla guancha Beaucournu, Alcover & Launay
Stenischia chini Xie & Lin
Stenischia liae Xie & Lin
Stenischia liui Xie & Lin
Stenischia wui Xie & Lin
Lagaropsylla duodecima Beaucournu & Kock
Jellisonia amadoi Ponce-Ulloa
Jellisonia mexicana Ponce-Ulloa
Plocopsylla crypta Beaucournu & Gallardo
Rowleyella gongshanensis Gong & Duan

1990

Hectopsylla pascuali Beaucournu & Alcover
Plocopsylla consobrina Beaucournu & Alcover
Agastopsylla boxi gibbosa Beaucournu & Alcover
Chiliopsylla allophyla tonnii Beaucournu & Alcover
Geusibia digitiforma Gong & Lin
Rhadinopsylla (Sinorhadinopsylla) leii Xie, Gong & Duan
Ceratophyllus (Emmareus) liae Wu & Li
Smitipsylla quadrata Xie & Li
Rowleyella nujiangensis Lin & Xie
Macrostylophora bispiniforma gongshanensis Gong & Xie
Hystrichopsylla llorentei Ayala & Morales

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Jellisonia guerrerensis Morales
Ctenoparia intermedia Beaucournu & Kelt
Plocopsylla wilesi Beaucournu & Kelt
Plocopsylla silewi Beaucournu & Kelt
Tiamastus gallardoi Beaucournu & Kelt
Liuopsylla clavula Xie & Duan
Nosopsyllus wualis rongjiangensis Li & Huang
Nosopsyllus wualis bosensis Li & Pan
Nosopsyllus wualis diandongensis Li, Xie & Yang
Nosopsyllus elongatus longchuanensis Li, Xie & Wu
Nosopsyllus elongatus puerensis Li, Xie & Yu
Nosopsyllus elongatus yanshanensis Li, Xie & Liao
Coptopsylla lamellifer tarimensis Yu, Ye & Cai
Vermipsylla yeae Yu & Li
Stenoponia meridiana Cao, Xie & Yu
Ctenophyllus (Ctenophyllus) armatus altaicus Yu
Ophthalmopsylla (Ophthalmopsylla) volgensis wuqiaensis Yu, Shen & Ye
Callopsylla (Callopsylla) yui Ye & Jiang
Leptopsylla (Leptopsylla) algira vogeli Beaucournu

1991

Xenopsylla cuisancei Beaucournu & Morel
Parapsyllus senellarti Beaucournu & Rodhain



LITERATURE ON SIPHONAPTERA

(1985-1986)

1985 (List 12)

Lin Z.-h. & Liao H.-r. Deformity of the genal comb in *Peromyscopsylla himalaica sinica* Li & Wang (Siphonaptera: Leptopsyllidae). Wuyi Sci. J. 5: 81-82.

1986 (List 10)

Li L. & Chen N.-h. Preliminary survey of rodent-like animals and their

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(1986-1988)

ectoparasites of the Mei-Hua Mountain Nature Reserve [Fujian] and adjacent area. Wuyi Sci. J. 6: 221-227.

Stojanovic, R., A. Gligic, M. Obradovic, R. Knezevic, B. Antonijevic, D. Drndarevic, D. Hlaca, J. Gaon & A. Arnautovic. Occurrence of rodents and their ectoparasites in natural foci of haemorrhagic fever with renal syndrome in Yugoslavia. Giorn. Malat. Infet. Parassit. 38(6): 685-687.

1987 (List 8)

Almeida, A. M. P. de, D. P. Brasil, F. G. Carvalho & C. R. de Almeida. Detection of *Yersinia pestis* in rodents and other small mammals in the plague foci of northeastern Brazil during the period from 1966 to 1982. Revta Saúde Públ. 21(3): 265-267.

Castro D. del C., R. Mauri, A. C. Cicchino & S. Mosquera. Ectoparasites of rodents from Buenos Aires province, Argentina (Acarina, Mallophaga and Suctoria). Revta Soc. Ent. Argent. 1985 [1987] 44(3-4): 317-327.

Gelker, F. Fleas on animals: prevention and control. Hassadeh 67(11): 2368.

Gills, D. E., A. J. Shepherd, P. A. Leman & B. H. Erasmus. Plague surveillance in the northern Cape province, South Africa. S. Afr. J. Sci. 83(3): 159-162.

1988 (List 7)

Costantini, R. & A. Iori. The Aphaniptera of Italy: records from 1964 to date. Parassitologia (Roma) 30 (Supplement): 58-59.

Costantini, R., P. Lanfranchi, A. Iori, M. A. Abdullatif & T. Balbo. Contributions to the study of the Siphonaptera of Somolia. Parassitologia (Roma) 30(2-3): 203-209.

Gants, R. Choose your weapon against fleas and ticks. Vet. Econ. April, 76...95.

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(1988-1989)

- Martinez, F. A.** Arthropods of the wild fauna. I. Insects and arachnids Veter. Argent. 5(45): 411-417.
- Mazzini, M. A., M. I. Fridmanis, H. J. A. Abarrio & G. L. Carbajal.** Tungaisis. Arch. Argent. Dermat. 38(6): 403-408.
- Pizzol, J. L.** Incidence of dermatoses in children between 0 and 6 years old attending municipal nurseries of Viana, Espírito Santo, in 1985. Anais Brasil. Dermat. 63(1): 15-17.
- Renapurkar, D. M.** Distribution and susceptibility of *Xenopsylla astia* to DDT in Maharashtra State, India. Insect Sci. & Appl. 9(3): 377-380.
- Saupe, E.** Parasitic diseases of hedgehogs and their treatment. Prakt. Tierarzt 69(12): 49-54.
- World Health Organization.** Human plague in 1987. Weekly Epidemiol. Rec. 63(47): 360-362.

1989 (List 5)

- Anonymous.** Veterinary technician 1989 flea and tick product guide. Vet. Tech. 10(4): 253...270.
- Beaucournu, J.-C. & D. Kock.** Notes sur les Ischnopsyllinae du Continent Africain. II. Descriptions de *Lagaropsylla duodecima* n. sp., en provenance du Togo et de la femelle véritable de *L. leleupi* Smit, 1957. Senckenbergiana biol. 70(1/3): 77-82.
- Bei-Bienko, G. Ya (Ed.)** *Keys to the insects of the european part of the USSR.* Volume V. Diptera and Siphonaptera. Part I. Leiden, Netherlands, E. J. Brill. xxii-1233 pp. [Keys to the fauna of the USSR, No. 100.]
- Bei-Bienko, G. Ya (Ed.)** *Keys to the insects of the european part of the USSR.* Volume V. Diptera and Siphonaptera. Part II. Leiden,
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(1989)

Netherlands, E. J. Brill. xxi-1505 pp. [Keys to the fauna of the USSR, No. 103.]

Braack, L. E. O. Arthropod inhabitants of a tropical cave "island" environment provisioned by bats. Biol. Conserv. 48(2): 77-84.

Burdelov, L. A., I. Zh. Zhubanazarov & N. F. Rudenchick. The size of small mammals and abundance of their fleas. Med. Parazit. Parazitar. Bolezni (4): 42-45.

Ershova, L. S., O. S. Serzhanov, A. P. Kogerman, S. I. Zolotova & A. F. Tkachenko. Effect of a synthetic analogue of the juvenile hormone on *Xenopsylla skrjabini*. Med. Parazit. Parzitar. Bolezni (5): 85-87.

Fain, A. & R. W. Mason. A new heteromorphic deutonymph (Acari: Winterschmidtidae) phoretic on the flea *Pygiopsylla hoplia* Jordan & Rothschild in Australia. Aust. ent. Mag. 16(2): 43-46.

George, R. S. Some bird flea records from Britain (Siphonaptera). Entomologist's Gaz. 40(4): 314.

Gómez, M. S., M. Pujols & J. Gállego. On the parasitocoenosis of fleas on *Mus musculus* (Rodentia: Muridae) and *Crocidura russula* (Insectivora: Soricidae) in the Ebro delta: annual study. Revta Ibér. Parasitol. 48(3): 323-333.

Gruschwitz, W. Flea records from the Stassfurt area (Insecta: Siphonaptera). Ent. Nachr. Ber. 33(3): 138-139.

Kilonzo, B. S. & I. R. S. Khama. The effects of goat (*Capra hircus*) age and sex on flea infestation in Tanzania. Bull. Anim. Hlth Prod. Afr. 37(1): 61-66.

Michel, P., L. le Jan, D. Candito, J. C. Mouden & P. Coulanges. Plague: an epidemiological reality in Madagascar in 1989. Medecine et Armees 17(5): 373-377.

(1989-1990)

Parer, I. & T. J. Korn. Seasonal incidence of myxomatosis in New South Wales. Aust. Wildl. Res. 16(5): 563-568.

Redgate, N. D. *Ischnopsyllus octactenus* (Kolenati) (Siphonaptera: Ischnopsyllidae): a new vice county record. Entomologist's Gaz. 40(4): 336.

Ribeiro, H. & R. Capela. Fleas of Portugal (Insecta: Siphonaptera). IV. A new record: *Odontopsyllus quirosi quirosi* (Gil Collado, 1934). Bolm Soc. port. Ent. No. 110 IV(8): 98-99.

Sleeman, P. *Stoats & Weasels, Polecats & Martens.* Whittet Books Ltd., London. 119 pp. [Siph.: 9, 26, 69, 80-82.]

Stanko, M. Ectoparasites (Insecta: Siphonaptera) of small mammals (Insectivora, Rodentia) of the Zoological Gardens in Košice. Zborník Východoslov. múzea v Košiciachm Prír. Vedy 30: 83-95.

World Health Organization. Human plague in 1988. Weekly Epidem. Rec. 64(45): 345-347.

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It will be understood that - although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in

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the USSR or China, are in Russian or Chinese (but often with summaries in English).

"1 March is particularly connected with fleas in Sussex. The saying is, 'On the first of March, the fleas start jumping.' In West Sussex, we say 'Never open your windows, or you will get a swarm of fleas in the house.' In East Sussex, they say, 'Always open your windows, or you will get a swarm of fleas in the house.' Another version is, 'If from the fleas you would be free, on the first of March let all your doors and windows open be.'

"This comes from West Sussex. In East Sussex, the final words are 'closed be'.

"Whether you believe in the East or West Sussex admonitions, 1 March seems to be the day to start spring cleaning, by sweeping out the cracks of the doors and windows. No doubt fleas were once a very great 'irritation' to the good Sussex housewives."

(from: Chapter 2, Fleas, Cuckoos and Cocks, in *A Sussex Garland, a lighthearted collection of rhymes, recollections and recipes of the Sussex year*, by Tony Wales. Published by Countryside Books. Newbury, Berkshire. Date unknown to us. Courtesy of Richard Lane, London School of Hygiene and Tropical Medicine.)

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